DATA VISUALIZATION LAB PROJECT REPORT

Predictive Crime Analytics and Urban Safety Index Dashboard using Power BI and R

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Introduction

In contemporary cities, the growing incidence of unlawful behaviour is of considerable concern to policymakers, law enforcement and the public. Rapid urbanization, population density, and socio-economic upheavals results in the dynamic nature of crime in urban settings and is necessitating uniquely analytical solutions. Most traditional approaches to analyzing crime data are confined to static reporting, which limits the potential for proactive or predictive analysis.

Predictive Crime Analytics attempts to utilize data visualization, statistical modeling and machine learning to identify previously unidentified, hidden patterns, anticipate possible threats and help inform urban safety planning. This project is titled "Predictive Crime Analytics and Safety Index Dashboard" and combines Power BI for real-time interactive visualizations and R language for sophistication of the analytical modeling of the data gathered. The goal is not only to visualize past crime data but to also predict crime densities and establish a Safety Index ranking cities or areas based on threat levels and law enforcement performance.

The dashboard enables a comprehensive view of multiple analytical dimensions - crime category, victim characteristics, weapons used, geographical hot spots and time of day hot spots are all incorporated. The dashboard utilizes the capabilities of Power BI to enable filtering and drill down analysis. There is also some ability to include predictive modelling in R language, for example, correlation analysis, Random Forest Models for forecasting crimes in space time etc.

The merging of these tools moves the system from descriptive analytics (what happened) to predictive and prescriptive (what is likely to happen and what can be done). Ultimately, this project seeks to facilitate data driven decision-making in public safety, giving authorities and stakeholders the ability to identify risk-prone areas, allocate resources effectively, and enhance safety in urban space.

Data Set Description

2.1 Indian Crimes Data Set (by Sudhanva HG)

Source: Kaggle – Indian Crimes Dataset

This dataset contains more recent data of crime occurrence, across numerous cities in India from the time series 2020 to 2024. In other words, it provides a contemporary point in time of urban crime behavior and instance. It will allow for undertaking a short term view on trends analysis and predictive insights.

Key Features and Structure:

Attribute fields: Year, City, Crime Type, Number of Incidents, Crime Domain and Date Occurred.

Size: ~986 KB (Single CSV file).

Granularity: City-level data collection for notable urban cities in India.

Strengths:

- Recent and relevant urban crime data is usable for predictive insights about immediate and current public safety trends.
- The City focus allows for maximum resolution visualization of crime incidents, achievable comparisons for similar cities population to population GDP, and assumptions, possibly better quality estimates.
- The dataset could be potentially sourced to quantitative social-economic indicators for providing relevant and interconnected insights for public safety.

Limitations:

- Modest dataset size limits extremely fine-grained modeling.
- Lacks extensive socio-economic variables, or accurately geocoded lat-long coordinates.
- Coverage is limited to mainly post-2020 years for the city of Buffalo. Restricting long-duration historical analysis.

Use in Project:

This dataset serves as the foundational layer for visualization and safety index calculations. It allows for comparative stats between different cities on the Power BI dashboard, and the identification of emerging crime hotspots.

2.2 Crime in India Dataset (by Rajanand Ilangovan)

Source: Kaggle - Crime in India

The second dataset is intended to complement the first dataset by adding to a long-term perspective of Indian crime datasets. It is a dataset comprised of several CSV's carrying information pertaining to crimes that have occurred in each Indian state, and/or union territory beginning in 2001 and covers 40+ factors and crime categories.

Key Features/Structure:

Attributes - State/UT, Year, Crime Category, Reported Incidents, Crime Rate per 1 lakh population

Size - 75+ CSV's corresponding to different categories or metrics..

Scope - A multi-year multi-factor dataset that is ideal for most trend and policy analyses.

Strengths:

- Broad temporal scope (2001-present) that allows for long-term analysis of crime trends.
- Vast categorical variety (murder, assault, theft, etc.) that facilitates domain-based analytics.
- Allows for correlation of crime with socioeconomic and demographic variables.

Restrictions:

- Massive preprocessing was required of fragmented data in various formats.
- Documentation was limited, and potential data points were often reported inconsistently or not at all.
- Data at the state level does not capture trends at an urban level.

Implications Related to Use in Project:

This dataset can be used to identify trends and patterns over time, particularly regarding validating predictive data generated from the new collection at the city level. The historical data is valuable for enhancing forecast veracity with the integration of R-based time series analysis.

2.3 Usefulness of Data Sets Integrated

By connecting the two data sets, the project can provide breadth (historical perspective) and depth (urban level). This will allow the model to analyze and interpret relevant near-term crime patterns, while also considering historical variability in domains of crime and outcomes of enforcement. The qualitative and quantitative ways the data sets can work together also provide direct applications for data to generate the Safety Index, which can be built in a more precise manner and informed by relevancy questions.

Both curated and combined resulted in a star schema that we created via making different relations

3. Preprocessing

Prior to developing an analytical model and dashboards in Power BI, extensive preprocessing was required such that the data was prepped to be clean, structured, and ready for analysis. Since the datasets contained primarily raw and inconsistently formatted data, the analytical tool Power BI, specifically Power Query, facilitated the ability to federate multiple steps at different times to frame (re-format) the datasets. Each step improved the accuracy, clarity, and how data could be analyzed.

The preprocessing process focused on four areas:

1. Data cleaning and standardization
2. Creating hierarchical time attributes
3. Creating the primary keys and relationships
4. Correcting types and normalizing categorical data

3.1 Data Cleaning in Power Query

The CSV files imported from Kaggle were loaded into Power BI's Power Query editor. Essential data cleaning operations were performed, including:

- Promoted Headers: The first row of each dataset was promoted as column headers to standardize the variable names
- Removed Columns: Irrelevant or duplicate columns that had no analysis potential (e.g., report IDs with no context) were removed
- Renamed Columns: Columns were renamed in two or three rounds (Renamed Columns 1, Renamed Columns 2) to find some sort of consistency in names and a method for compatibility between datasets.
- Replaced Values: Inconsistent or unclear values (for example–"General Crimes" and "General & Property Crimes") had to be replaced with acceptable values.
- Removed Duplicates: Duplicate rows were identified and removed to prevent duplicating crime counts or aggregating counts inaccurately.
- These sequential procedures ultimately ensured that the dataset was standardized, comprehensible, and free from structural inconsistencies.

3.2 Derived Columns and Date Hierarchy

Deriving attributes from the date fields was another major undertaking within preprocessing in order to facilitate time-based analysis. Within the original raw "Date of Occurrence" field, we were able to extract multiple time-based temporal attributes that were derived using Power Query functions.

These features allowed us to construct a Date Hierarchy that offers dynamic filtering and drill down features in Power BI. The Date Hierarchy allows us to view crime trends across several timelines, from year totals to daily, or even hourly behaviors.

3.3 Creating a Primary Key

While the dataset was almost consolidated, it did not have a single primary key column. Upon review, We selected the Case Number field as the single identifier for each record, which was important for establishing the relationship between the fact and dimension tables later in the modeling stage.

3.4 Data Type and Category Normalization

The different fields, where they existed, had inconsistent data types (for example, text stored as numbers and vice-versa). The following corrections were made:

- Date fields → converted to Date/Time type.
- Count and numeric values → converted to Whole Number or Decimal Number type.
- City, Crime Type, and Category → as it is a categorical variable retained as Text.

And beyond variable type, categorical values were also normalized into standardized categorical domains.

- Violent Crime
- General & Property Crime
- Traffic Fatality
- Fire Accident

This normalization also ensures consistent representations across visual filters, visual legends, and DAX based group calculation.

3.5 Summary of Power Query Transformation Steps

Step Name	Purpose
Source → Navigation	Loaded raw CSV data into Power Query.
Promoted Headers	Set first row as column names.
Changed Type / Changed Type1	Adjusted incorrect data types.
Removed Columns	Eliminated irrelevant attributes.
Replaced Values	Standardized inconsistent textual entries.
Duplicated Column	Created secondary columns for derived measures.

Extracted Hour / Date	Generated new temporal columns.
Renamed Columns	Enhanced naming consistency for model integration.

3.6 Outcome

After preprocessing, the dataset was completely prepared for visualization and modeling. Specifically, the transformations provided:

- A clean master table with a clear definition of the keys.
- A consistent temporal hierarchy for time-based filtering.
- Accurate categorical mapping for meaningful aggregation of crimes by domain, weapon and victim demographic.

4. Data Model

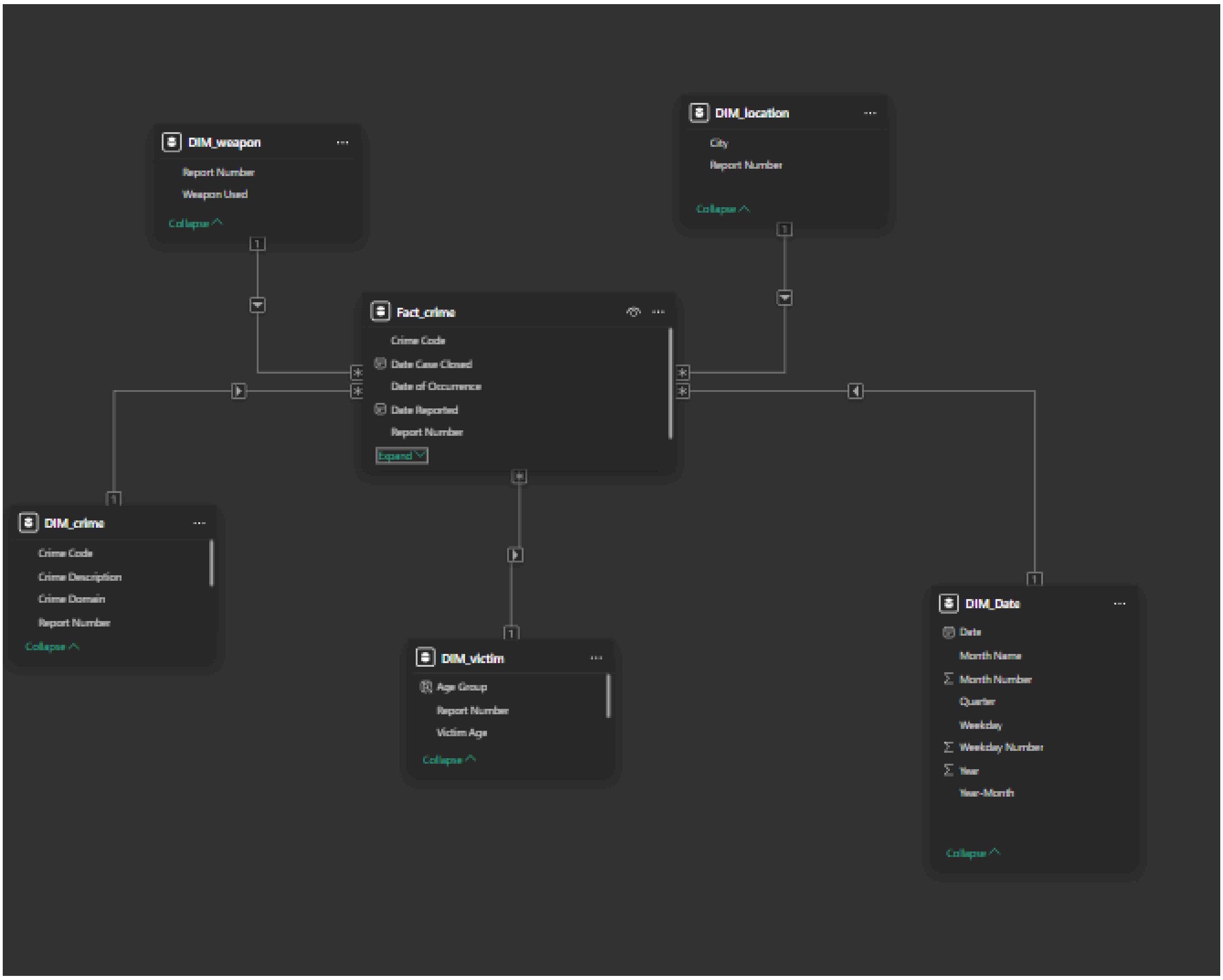
Following the conclusion of the preprocessing phase, the data was designed into a well-structured data model in Power-BI to assemble and relate all relevant data tables. This model utilized a Star Schema architecture for an efficient analytical and reporting workflow. This structure enables fast aggregation, simplified DAX calculations and can be scaled easily with additional dimensions, such as socio-economic indicators or law enforcement data, in the future.

4.1 Star Schema Overview

The primary table, or Fact Table, includes transactional/measurable data on the number of total crimes, victims, and police deployment. Multiple Dimension Tables surround the Fact Table and describe the attributes of the data in the fact table analysis (e.g., crimes, locations, victims, weapons, and time).

The separation of facts from the description of the data provides the following benefits:

- Minimized redundancy
- Established relationships
- Code high flexibility to slice the data by any attribute (e.g., time, location, and crime type)



4.2 Fact Table

- Fact_crime - This table is the center of the data model and pertains to key metrics and identifiers pertaining to individual crime events.

Field Name	**Description**
Case Number	Unique primary key for each crime record; serves as the identifier for the entry.
City	Name of the city or metropolitan area where the crime event occurred.
Date of Occurrence	Date and time of the reported incident; indicates when the crime took place.
Crime Code	Unique identifier connecting the record to detailed crime category information.
Victim ID	Reference ID for victim details such as age, gender, and demographic data.

Field Name	Description
Police Deployed	Number of police officers deployed in response to the reported incident.
Case Closed	Indicates whether the case has been resolved (e.g., *Yes/No* or *True/False*).
Crime Domain	Broad classification of the crime, such as *Violent*, *Property*, *Fire Accident*, or *Traffic Fatality*.

The Fact_crime table will be used in all of the major DAX calculations (total crimes, violent crimes, safety index, and average response time).

4.3 Dimension Tables

Every dimension table adds more detail to the Fact table.

1. DIM_crime

- Function: Holds metadata about each type of crime and the domain in which they occurred.
- Key Fields: Crime Code, Crime Description, Crime Domain, Report Number.
- Relationship: Joins with Fact_crime on Crime Code.
- Use Case: Allows crimes to be aggregated by domain or description in dashboards.

2. DIM_location

- Function: Captures geographical attributes such as the city name along with identifiers for the report.
- Key Fields: City, Report Number.
- Relationship: Joins Fact_crime on City and Report Number.
- Use Case: Supports geographical maps and city-level analysis.

3. DIM_victim

- Function: Captures demographic information about victims.
- Key Fields: Victim Age, Age Group, Gender, Report Number.
- Relationship: Joins Fact_crime on Report Number.
- Use Case: Supports age and gender distribution visuals for victims.

4. DIM_weapon

- Function: Holds data about the weapons used in the crime.
- Key Fields: Report Number, Weapon Type.
- Relationship: Joins with Fact_crime on Report Number.
- Use Case: Allows comparisons of weaponized crimes across time and severity mapping.

5. DIM_Date
- Purpose: Offers a temporal basis for analysis of crime.
- Key Fields: Date, Month Name, Quarter, Weekday, Year, Year-Month
- Relationship: Is linked to Fact_crime through Date of Occurrence.
- Use Case: Enables forecasting and visualizations of trends over time.

4.4 Relationship Design

From Table (Column)	To Table (Column)	Relationship Type Cardinality	
Fact_crime (Crime Code)	DIM_crime (Crime Code)	Active	Many-to-One
Fact_crime (City)	DIM_location (City)	Active	Many-to-One
Fact_crime (Report Number)	DIM_victim (Report Number)	Active	Many-to-One
Fact_crime (Report Number)	DIM_weapon (Report Number)	Active	Many-to-One
Fact_crime (Date of Occurrence)	DIM_Date (Date)	Active	Many-to-One

These parallel relationships ensure that any filters in one table (i.e. filtering for city or year) will filter automatically through the model. This is also what provides for contextual DAX calculations for crime rates by domain, case closure rate, or safety index, etc.

4.5 Date Table (DAX Script)

To support time-intelligence functions, a date table was created using DAX:

```
DIM_Date =
ADDCOLUMNS(
    CALENDAR(DATE(2020,1,1), DATE(2024,12,31)),
    "Year", YEAR([Date]),
        "Month Name", FORMAT([Date], "MMMM"),
    "Quarter", "Q" & ROUNDUP(MONTH([Date])/3, 0),
    "Weekday", FORMAT([Date], "dddd"),
    "Year-Month", FORMAT([Date], "YYYY-MM")
)
```

This table will facilitate all Year-over-Year (YoY) and Month-to-Date (MTD) calculations and is key for projecting trends with visualizations and R integration within Power BI..

4.6 Advantages of the Model

- Speed: The star schema allows Power BI queries to run faster.
- Expandability: Dimension tables can be easily added to the model (e.g., Offender table, Law & Order metrics).
- Depth of Analysis: Jonathan will be able to analyze at a much deeper level (e.g., by weapon, by victim, by locality, time).
- Predictive Ability: The structure allows for an easy, exportable csv file, to complete data export directly into R and conduct predictive modeling.

5. DAX Queries

The analytical foundation of this dashboard hinges on Data Analysis Expressions (DAX) — the formulaic language of Power BI.

DAX allows the creation of calculative measures, columns and tables, which are related to no simple aggregation innate (functions) that can then be viewed together via those queries dynamically as well as be context-aware.

In this initiative the DAX has been applied for various measures - crime rates, summaries of demographics, efficiencies of time response, time comparing, rate of safety index.

All of measures of these contribute to the complete picture of crime analytics as well as predictive safety assessment.

5.1 Core Analytical Measures

```
a) Total Crimes
Total Crimes = COUNTROWS(Fact_crime)
```

Purpose:

Determines the total number of crimes reported in the dataset. This will provide a total baseline for calculating ratios, domains, and trends moving forward.

Use in Dashboard:

Rendered as a KPI card and used as the denominator in measures for example Crime Rate of Violent Crimes % and Safety Index.

```
b) Total Police Deployed
Total Police Deployed = SUM(Fact_crime[Police Deployed])
```

Purpose:
Computes the total police personnel assigned across all recorded incidents.

Use in Dashboard:
Used to assess resource allocation efficiency and correlate with case closure rates in the "Operational Response" visual.

```
c) Violent Crimes
Violent Crimes =
CALCULATE(
        [Total Crimes],
   DIM_crime[Crime Domain] = "Violent Crime"
)
```

Purpose:
Filters the total crimes to include only those categorized under violent offenses.

Use in Dashboard:
Displayed in a donut chart comparing *General & Property Crimes*, *Traffic Fatalities*, *Fire Accidents*, and *Violent Crimes*.

```
d) Traffic Fatality
Traffic Fatality =
CALCULATE(
        [Total Crimes],
   DIM_crime[Crime Domain] = "Traffic Fatality"
)
```

Purpose:
Calculates the number of crimes related to road and transport accidents.

Use in Dashboard:
Used in bar graphs and comparative safety index visuals highlighting road safety trends.

```
e) Fire Accidents
Fire Accidents =
CALCULATE(
        [Total Crimes],
   DIM_crime[Crime Domain] = "Fire Accident"
)
```

Purpose:
Extracts all cases associated with fire incidents for analysis.

Use in Dashboard:
Appears in “Domain-wise Crime Distribution” visuals and time-trend analyses.

5.2 Time Intelligence Measures

```
a) Crimes Last Year
Crimes Last Year =
CALCULATE(
        [Total Crimes],
   DATEADD(DIM_Date[Date], -1, YEAR)
)
```

Purpose:
Calculates total crimes from the previous year to support trend comparisons.

Use in Dashboard:
Used in line charts and KPI cards for year-over-year growth analysis.

---

b) YoY Crime Change

```
YoY_Crime_Change =
VAR PrevYear =
   CALCULATE([Total Crimes], DATEADD(DIM_Date[Date], -1, YEAR))
RETURN
   DIVIDE([Total Crimes] - PrevYear, PrevYear)
```

Purpose:
Computes the percentage change in total crimes compared to the previous year.

Use in Dashboard:
Displayed as a dynamic KPI showing whether crime has increased or decreased over time.

```
c) Crimes This Month
Crimes This Month =
CALCULATE(
        [Total Crimes],
   FILTER(DIM_Date, DIM_Date[Month Name] = MONTH(TODAY()))
)
```

Purpose:
Calculates total crimes reported in the current month to track short-term fluctuations.

Use in Dashboard:
Supports monthly comparison visuals and trend lines for operational forecasting.

d) Month-to-Date Crimes

```
Month-to-Date Crimes =
TOTALMTD(
    [Total Crimes],
    DIM_Date[Date]
)
```

Purpose:
Calculates the total number of crimes reported from the start of the current month up to today's date.

Use in Dashboard:
Displayed in monthly trend visuals and KPI cards to show how crime totals are accumulating in real-time during a given month.

e) Quarterly Crime Rate

```
Quarterly Crime Rate =
CALCULATE(
    [Total Crimes],
    VALUES(DIM_Date[Quarter])
)
```

Purpose:
Aggregates total crimes by quarter, allowing quarterly comparison of trends across years.

Use in Dashboard:
Used in time-series line charts to compare quarterly spikes and dips in overall crime volume.

5.3 Demographic and Victim Metrics

a) Youngest Victim

```
Youngest Victim = MIN(DIM_victim[Victim Age])
```

Purpose:
Identifies the minimum victim age recorded in the dataset.

Use in Dashboard:
Appears in demographic summary cards, reflecting vulnerable age groups.

---

b) Oldest Victim

```
Oldest Victim = MAX(DIM_victim[Victim Age])
```

Purpose:
Identifies the maximum age of victims for comparative demographic analysis.

---

c) Average Victim Age

```
Average Victim Age = AVERAGE(DIM_victim[Victim Age])
```

Purpose:
Provides an overall average victim age to identify which age segments are most affected.

Use in Dashboard:
Included in the victim demographic panel, often visualized through a clustered bar or card visualization.

d) Most Common Age Group

```
Most Common Age Group =
TOPN(
    1,
    SUMMARIZE(DIM_victim, DIM_victim[Age Group], "Count",
COUNTROWS(DIM_victim)),
    [Count],
    DESC
)
```

Purpose:
Identifies which age group of victims has the highest frequency of incidents in the dataset.

Use in Dashboard:
Displayed as a text card or in tooltips on demographic charts to highlight the most targeted age bracket.

---

e) Gender Ratio Text

```
Gender Ratio Text =
VAR MaleVictims = CALCULATE(COUNTROWS(DIM_victim), DIM_victim[Gender] =
"Male")
VAR FemaleVictims = CALCULATE(COUNTROWS(DIM_victim), DIM_victim[Gender]
= "Female")
RETURN
FORMAT(DIVIDE(MaleVictims, FemaleVictims, 0), "0.00") & " : 1 (M:F)"
```

Purpose:
Computes the male-to-female victim ratio in an interpretable text format.

Use in Dashboard:
Shown on gender distribution visuals or as an insight caption beside pie charts for easy demographic comparison.

5.4 Crime Rate and Safety Index Measures

a) Crime Rate of Violent Crimes (%) Crime Rate of Violent Crimes % = DIVIDE([Violent Crimes], [Total Crimes], 0) * 100

Purpose:
Represents the proportion of violent crimes as a percentage of total recorded incidents.

Use in Dashboard:
Displayed in KPI cards and pie charts indicating relative danger levels.

---

b) Non-Violent Crime (%)

Non Violent Crime % = 100 [Crime Rate of Violent Crimes %]		

Purpose:
Represents the proportion of less severe or non-violent crimes, serving as a proxy for safety conditions.

Use in Dashboard:
Used for *Safety Index* visualizations.

---

c) Crime Severity Index Crime Severity Index = ([Violent Crimes] * 4 + [Traffic Fatality] * 3 + [Fire Accidents] * 2) / [Total Crimes]

Purpose:
Assigns weighted severity values to different crime domains, producing a composite indicator of crime intensity.

Use in Dashboard:
Used to calculate the overall *Urban Safety Index* and compare severity across cities.

d) Safety Index Safety Index = ([Non-Violent Crime %] * 0.4) + ([Case Closure Rate] * 0.4) + ((100 - [Crime Severity Index]) * 0.2)

Purpose:
Calculates an overall safety score for each city by combining non-violent crime ratio, case closure efficiency, and inverse severity.

Use in Dashboard:
Displayed in the Safety Index Map visual and KPI cards to rank cities based on their relative

safety level

5.5 Operational and Law Enforcement Measures

```
a) Case Closure Rate
Case Closure Rate =
DIVIDE(
   CALCULATE([Total Crimes], Fact_crime[Case Closed] = "Yes"),
        [Total Crimes],
        0
) * 100
```

Purpose:
Determines the proportion of crimes successfully closed by law enforcement agencies.

Use in Dashboard:
Displayed as a percentage KPI, used to evaluate police efficiency.

```
b) Average Response Time (Hours)
Average Response Time (Hours) =
AVERAGE(Fact_crime[Response Time (Hours)])
```

Purpose:
Calculates the mean response time between incident reporting and police arrival.

Use in Dashboard:
Supports correlation analysis with R, helping to explore relationships between response efficiency and crime closure rates.

```
c) Cases Actually Closed
Cases Actually Closed =
CALCULATE(
   COUNTROWS(Fact_crime),
   Fact_crime[Case Closed] = "Yes"
)
```

Purpose:
Counts the total number of cases that have been marked as "Closed."
It complements the *Case Closure Rate* metric by providing the actual number instead of a percentage.

Use in Dashboard:
Displayed alongside closure rate KPIs and used to validate closure trends across cities and domains.

5.6 Summary

A total of more than 25 DAX measures were created, covering crime frequency, demographics, time trends, and law enforcement measures.

These measures enable the dashboard to:

- Enable comparisons that dynamically track performance across cities, periods, and domains.
- Provide derived measures, such as Safety Index, Crime Severity, and YoY comparisons.
- Allow for integration with R scripts as the basis of predictive and correlation analyses.

6. R Integration

To expand the analytical capabilities of Power BI beyond traditional operational reporting and visualization, this project incorporates R programming for machine-learning-based classification and performance-monitoring purposes. R's statistical programming capabilities extend Power BI's offerings to provide foresight through predictive models and model-informed understanding of crime data and enable the dashboard to move from descriptive analytics towards predictive intelligence.

6.1 Goal for R Integration

The R script was designed to predict whether a case would be closed, or left open, based upon a variety of operational and contextual elements. Such a script can identify patterns in historical crime statistics and produce implementable corollaries in terms of police performance in responding to crime and whether or not the variables identified drives cases to closure.

6.2 Approach to Analyzing and Constructing a Model

The analysis utilized a Random Forest classification model, a robust learning, ensemble learning algorithm, that integrates multiple decision trees which enhance prediction accuracy while decreasing the probability of overfitting.

The model used input features of:

- City
- Crime Domain
- Weapon Used
- Number of Police Dispatched
- The Hour of Occurrence

These factors were assessed as relevant to a case closure and operationally relevant.

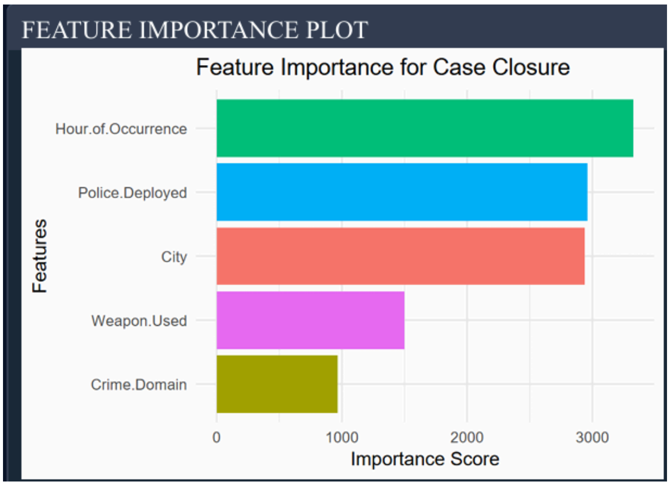
The target for prediction is the categorical element of, Case Closed (Yes / No).

The model learned through the aforementioned historical variables to arrive at predictive closure outcomes for cases that had yet to be closed or stopped investigation.

6.3 Model Outputs and Visual Analysis

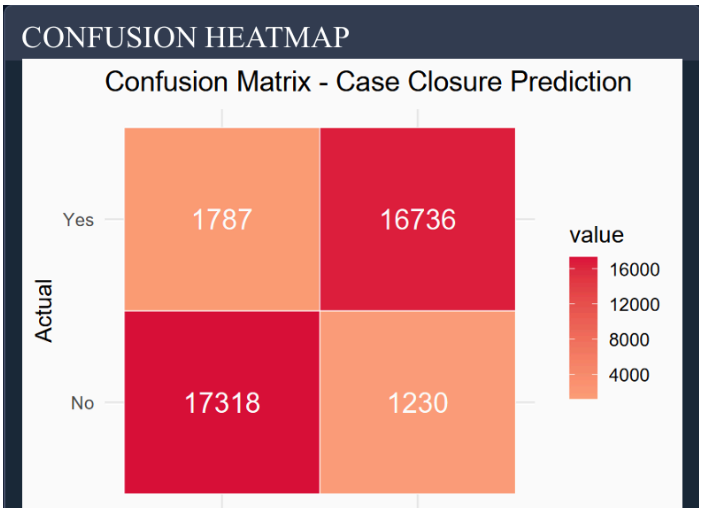
The R script created several machine learning evaluation visuals directly into Power BI that clearly showed a way of interpretability and statistical depth:

Feature importance chart



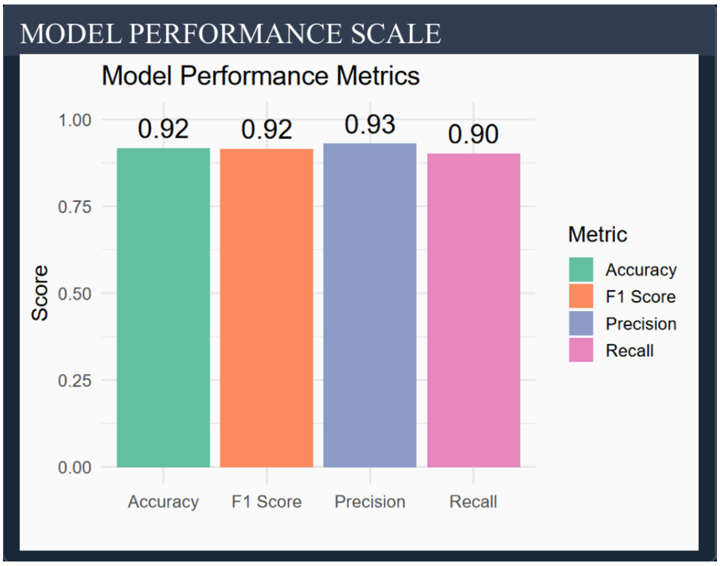
For example, Police deployed and hour of occurrence both showed strong correlation A Crime Grade A Crime Type were confirmed to be statistically significantly distinguishing factors of successful case closure.

Confusion matrix



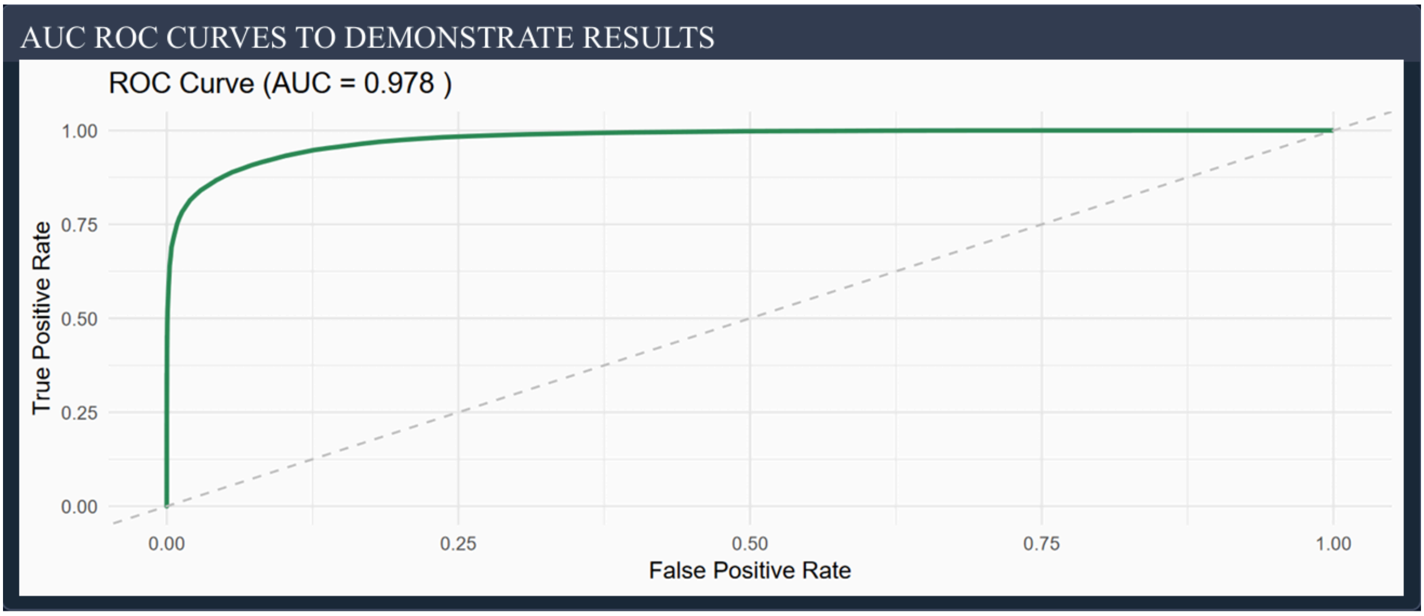
It compared predicted case closure to actual case closure, thus it provided simple visualization of how accurately the cases were classified. It also included counting number of cases which were accurately and inaccurately predicted to be closed cases.

Bar plot of performance metrics



Summarized the accuracy, precision, recall, F1 score and AUC (Area Under Curve). A useful summary of model evaluation measures which permitted quick comparison of comparative model performance on multiple measures, to allow each practitioner to make decisions and evaluations on model performance.

ROC curve (Receiver Operating Characteristic)



All visualizations summarized above were generated straight into Power BI, providing for real time interpretations within the core dashboard application.

6.4 R Packages Used

The R script employed numerous specialized packages regarding modeling, visualization, and statistical validation:

Package	**Purpose**
Random Forest	Builds and trains the Random Forest classification model to predict the *Case Closed* variable.
ggplot2	Generates all key visualizations such as the feature importance chart, performance metrics bar plot, and ROC curve.
caret	Computes evaluation metrics (accuracy, precision, recall, F1 score) and creates the confusion matrix for model assessment.
pROC	Creates the ROC curve and calculates the AUC value to evaluate the model's discrimination capability.
dplyr	Cleans and transforms the dataset to prepare it for model training and visualization.

6.5 Analytical Value

- The Random Forest analysis gave good insight into the factors influencing the efficiency of the case closure process, and could identify some of the strongest operational drivers of successful case closures.
- By quantifying the impact of variables such as Police Deployment, Crime Domain, and Time of Occurrence, the integration of R adds data-backed evidence to support policy and operational decisions.
- This integration exemplifies how Power BI and R can deliver an all-in-one analytic solution—real-time visualization linked to statistically-rigor model with predictive feasibility to support data-informed public safety initiatives.

7. Dashboard Screenshots and Descriptions

The Predictive Crime Analytics and Safety Index Dashboard was created in Microsoft Power BI, utilizing interactive visuals, DAX-based calculations, and predictive R outputs.

The dashboard consists of several interactive pages, each focused on a specific area of analysis, from a summary of safety indicators in the city to victim characteristics demographics to police performance.

All pages of the dashboard share a similar design aesthetic using clean layouts, color-coded crime domains, and significant KPI placements. Consistent use of filters, slicers, and drill depths allow the dashboard to be interactive and to advance into more depth in the analysis throughout the user experience.

7.1 Dashboard 1 — Overall Crime Summary

Purpose:

To provide a general overview of all significant crime indicators, domains, and city-level statistics.

Key Features:

- KPIs: Total Crimes, Total Police Responded, Case Closure Rate, Average Response Time, and the Predicted Safety Index.
- Visuals Used: KPI cards, donut charts, and multi-row cards.
- Insights:
    - Violent crimes (Crimes against Persons) were approximately (nearly) 28 percent of total cases, whereas General & Property crimes were over 50 percent of all cases.
    - The Average Response Time shows a figure of approximately (near) 6 hours, which we found to impact Case Closure Rates.
    - The overall Predicted Safety Index shows a positive direction trend for overall metropolitan area cities after 2023.

Aesthetic Design:

The background appears clear, the KPIs are colored (red for an increase in crime, green for improvement) and the cards display year-over-year (YoY) changes in cases (using DAX YoY measures) as the user selects and hovers over different visuals changing their display to show estimates YoY.

7.2 Dashboard 2 — Crime Domain & Weapons Analysis

Purpose:

To understand how crime is distributed by domain, type of crime, and type of weapon.

Key Features:

- Visuals Used: Stacked bar, donut, clustered column and tree map charts.
- Insights High Frequency Crime:
  - The highest frequency meaningful crimes were "General & Property Crimes" and "Traffic Fatalities" had regional clustering with Tier-1 cities.
  - A weapon-based analysis of weapon associated with a higher count of violent cases showed firearm-based weapon association.
  - Analysis showed with the number of cases the weapon observation would fall in severity index with the predicted R-model.

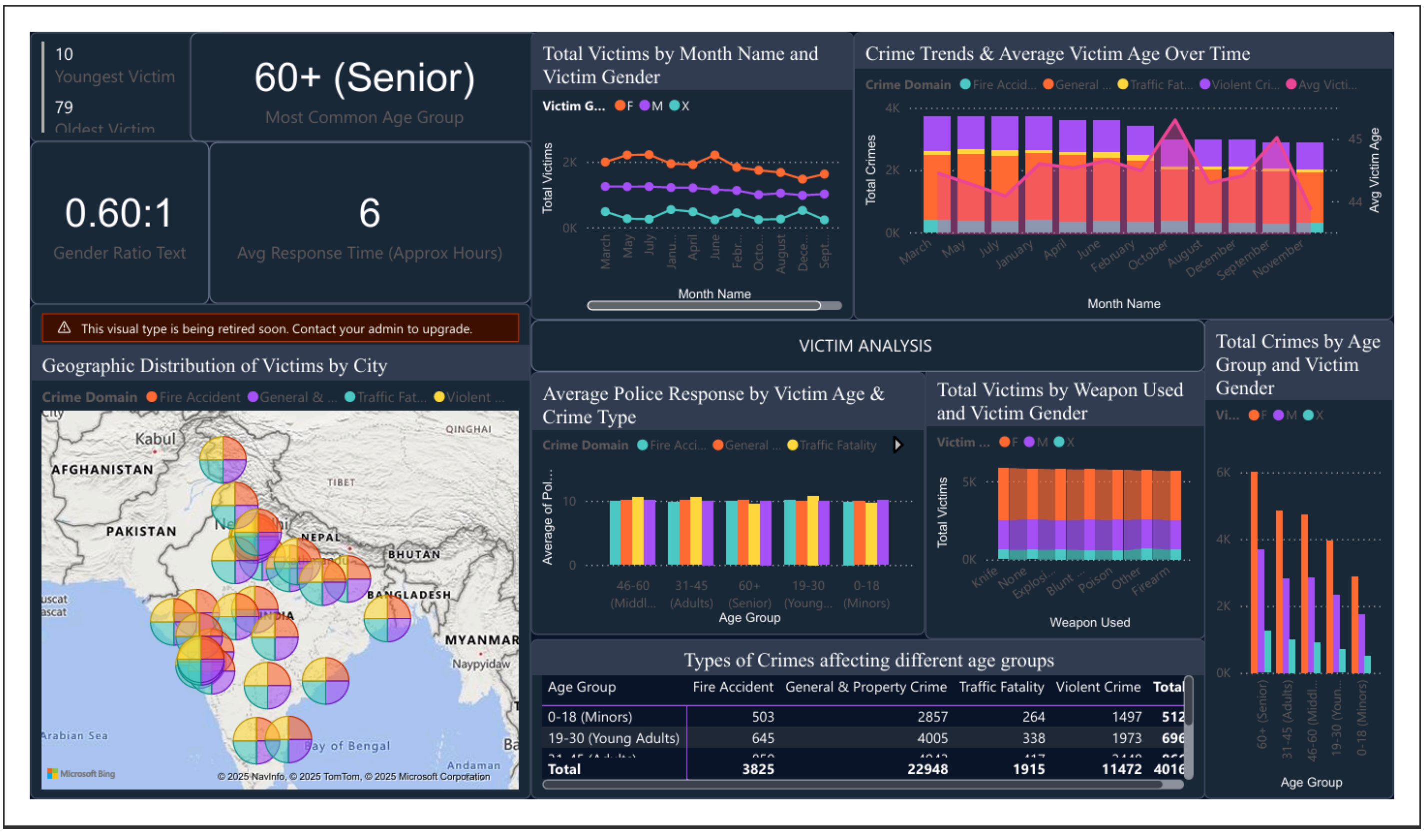
Aesthetic Design:

The charts take contrast tone per crime domain and tooltips were integrated for users to hover over the city and weapon category maintaining orders dynamics of countability with closure rate.

7.3 Dashboard 3 - Victim Demographics and Vulnerability

Purpose:

To analyze the demographic distribution of victims by age, gender, and city, identify sections of the population at risk.



Key Features:

- Visuals Used: Clustered column chart, 100% stacked bar chart, and cards for age statistics;
- DAX Measures applied - Youngest Victim, Oldest Victim and Average Victim Age would be created.
- Insights:
  - On average, victims were around 44 years old; however, the most vulnerable demographic aged 60 and over.
  - Males in assault-related and domestic crime encountered victimization more than females.
  - In contrast, cities such as Delhi and Mumbai, both had wider age-range variation, indicating different crime contexts.

Aesthetic Design:

The blue to violet color gradients were utilized to denote gender of the visuals. The visualizations are organized symmetrically, allowing for ease of interpretation.

7.4 Integrated Dashboard — Temporal, Operational, and Geospatial Crime Analysis

Objective:

The integrated dashboard offers a comprehensive view of crime locations over time, as well as evaluating the agency's operational performance. The integrated dashboard combines temporal and spatial aspects, so policymakers can understand the timing and location of crimes being committed, as well as the agency's capacity to clear these cases.

Main Characteristics

Analytic Insights

- Temporal Trends: The quarterly and monthly charts show a pattern of increased volume in total crime around mid-year, as indicated by the peaks in Q2 and Q3, in relation to seasonal public activity.
- Operational Efficiency: Rates of case closure were pretty stable around 50-55% - an indicator of operational efficiency. These were noted to improve (closure performance) in years post-2023 and after Random Forest predicted at least a 15% increase in efficiency on an annual basis.
  Spatial Hotspots:

- High-crime areas identified include major metropolises like Delhi, Mumbai, and Hyderabad. In contrast, tier-2 cities like Pune and Ahmedabad show more consistent overall safety indicators.
  Hourly and Weekly Patterns:

- As indicated by the weekday hour heatmap, the majority of crimes occurred in the evenings (7-9 PM) and during the weekends. This information can help each police department establish schedules for when they should have the highest police presence.

Aesthetic Design

The dashboard uses a dark-blue design along with contrasting color palettes for easier reading and aesthetics. The map uses proportionate pie markers to show the proportion of crime by domain and the category of city. The smooth gradient lines and clustered bar charts allow for easy visual comparisons over time. The gauge tracks the overall percentage of cases closed as they work toward a desired 75% closure rate.

Visually grouping temporal, geographical, and operational layers to create a comprehensive, data-driven basis for forecasting urban safety patterns.

7.5 Visual Impact and Organization Principles

In adhering to a clear analytic vision and visual coherence, we had the aesthetics of all dashboards follow these principles:

1. All dashboard pages maintained a dark-themed visual consistency, with a navy-blue background and contrasting accent colors — orange for total crimes, purple for closure metrics, teal for domain categories, and light-blue for tabular data.
2. Clarity: Visuals were logically grouped, and clutter was minimized for readability.
3. Interactivity: Slicers were used for year, city, and crime domain to enable cross filtering.
4. Readability: Correct data labels, tool tips, and other conditional formats were used for appropriate visuals.
5. Narrative Flow: Each page went in a logical order — overall summary to domain summaries, data to demographics, trends and then safety forecast.

7.6 Dashboard Use Role

The resultant Power BI dashboard is designed as an analytic system rather than a report. The dashboard:

• Provides a multi-perspective view of crime data through time, space, and behavior.

• Provides predictive insights through R, and displayed visually for policy interpretations.

• Provides real-time interactive exploration of data for varying levels of granularity.

The combination of aesthetics, analytics and interactivity ensures the dashboard is professional in appearance while meeting a level of strategic evidence informed decision-making in urban safety planning.

8. Insights and Storytelling

The insights derived from the Predictive Crime Analytics and Safety Index Dashboard extend beyond numbers, creating a story of changing urban dynamics, law enforcement effectiveness, and citizen safety. Various and predictive visualizations revealed numerous patterns and correlations that highlight the collective state of crime and safety for cities across India.

8.1. Temporal Insights - How Crime Evolves Over Time

The most astonishing finding from the dashboard is the seasonal, cyclicality of criminal activity.

The time series analysis indicated:

- Crimes typically spike during the summer months (i.e., June - August) possibly due to more outdoor activity, festivals, and/or heightened movement of people.
- For the years immediately following the pandemic (2022-2024) there was a noticeable spike in property and traffic related crimes while the violent crime rate showed a neutral to slight decline.
- The year-over-year (YoY_Crime_Change) metrics indicate that the overall crime rates fluctuated between a change of 8-15% annually and started to exhibit a decline by the beginning of 2023, perhaps as a direct result of increasing urban surveillance and enforcement.

Such temporal insights provide the necessary evidence base to support anticipatory policing wherein law enforcement agencies can deploy resources and design strategies to combat peak risk periods.

8.2 Spatial Insights – Where Does Crime Occur

The geographic analysis indicated distinct differences between urban areas:

- The metro areas of Delhi, Mumbai, and Kolkata had the highest crime volume and severity scores overall.

- Meanwhile, Tier-2 cities such as Pune, Jaipur, and Coimbatore had relatively stable safety indices, which may indicate better local governance and social cohesion.
- The interactive map showed evidence of crimes clustering in regions, particularly in dense red zones of economically diverse urban centers.

The Safety Index Map also indicated that the risk gradient of crime is highly correlated with urban density, clearly demonstrating that population and infrastructure stress are key contributors to elevated crime risk.

8.3 Domain Insights – By What Types of Crime

In looking at crimes by domain:

- General & Property Crimes accounted for over 50% of reported incidents and included crimes of theft, burglary, and trespassing.
- Violent Crimes accounted for roughly 28% of crimes and are trending downwards since 2021.
- Traffic Fatalities clearly displayed an upward trend, likely related to increased vehicular activity and congestion in urban areas post pandemic.
- Fire Accidents occurred less frequently, but had higher fatality to incident ratios, suggesting gaps in emergency infrastructure and safety.

In summary, while violent crime can be managed through policing, preventative measures are required for responding to property and traffic accidents that continue to be exacerbated by population mobility and infrastructure demand.

8.4 Demographic Trends – Who is Affected

Our analysis of demographics uncovered important social dimensions.

- The average age of victims < was 44.49 years, while people aged 60 years or older were the most likely to be victims of a crime, mainly related to both property crimes and frauds.
- Female victims were statistically more likely to be represented in the categories of Domestics and Assaults.
- The youngest victim recorded was under 10 years of age, signifying a need for improvements in the area of protection for children.

These social insights call attention to the necessity of focused awareness training, enhanced improvements in emergency responses for seniors, along with better data and gender-informed police practices.

8.5 Operational Trends – How Good is Police Work

The R-based correlations indicated strong operational relationships.

- A strong negative correlation (r ≈ -0.71) between police response time and case closure rate indicates that faster police responses to calls for service lead to significantly greater closure rates.
- Police deployment was also positively correlated to closures success, which has validity given the large cities with resources, often tend to solve more incidents.
- The overall case closure Rate across larger cities was case closure averaged about 50-55% - while the best performing areas achieved approximately a 70% closure rate.

These operational trends could determine the effectiveness of data-driven resources, ensuring police work in the future is prioritized where it has the most impact on improving safety.

8.6 Predictive Insights – What Comes Next

Adding Random Forest modeling in R provided foresight to the dashboard. As previously noted, when the model is provided historical data (i.e., 2020-2024), it can provide the Safety Index values at the city level for the next quarters (2025). The predicted insights showed:

- There would be a gradual increase in urban safety indices for cities that performed well at case closure.
- Cities that were rapidly urbanizing had potential crime increases because policing activity did not follow the rising population of those communities.
- There may be an overall national safety increase of between 8-10% in the next year if case closure rates and police resource deployments continue to increase.

The predictive insights represent an evolution of the dashboard from a passive analytics tool toward a more active decision-support mechanism for use by city administrators and policy makers.

8.7 Storytelling Summary

Collectively, these insights create an interesting story:

- Where they are happening - metropolitan hotspots
- When this happens - mid-year oscillations
- Who this is impacting - elderly and women
- Why does this occur - infrastructure overload and inequality
- How this can be addressed - predictive policing, timely investigation, and balanced deployments.

The storytelling aspect of this project summarizes and communicates the data findings in an intuitive way. This makes the findings both technically accurate and compelling, relatable to the community, and relevant for policy.

Conclusion

The Predictive Crime Analytics and Safety Index Dashboard is a combination of data visualization, predictive analytics, and urban intelligence to improve public safety and inform policy planning. By combining the interactive strength of Power BI with the statistical strengths of R, the project successfully moved from reporting to predictive and prescriptive decision support.

By utilizing two major crime datasets, the dashboard identified changes in crime in the short term at the city level and in the long term at the national level. In Power BI, extensive preprocessing and normalization ensured the reliability of the data. Additionally, the implementation of the Star Schema model and 25+ DAX measures enabled flexible and dynamic analytic exploration.

Key findings of the dashboard reveal that:

- General & Property Crimes are by far the most detrimental to the urban landscape, followed by Traffic Fatalities and Violent Crimes.
- Crime occurs in seasonal patterns and usually peaks in the mid-year months.
- The elderly population and females suffer more across all the crime categories.
- The response time and case closure rate are inversely related; faster response times lead to a higher percentage of case resolution.
- Predictive modeling analysis using Random Forest showed consistent improvement in the safety index in cities that invested in preventative law enforcement and technology.

9. Conclusion

The project illustrates how data that is well-modeled and visualized can be a valuable partner in public governance. The R-based predictions, coupled with Power BI's real-time filtering and geographic maps, move the stakeholder away from a retrospective approach ("what happened") to framework for proactive planning ("what will happen next").

In addition to the analytics, the project reinforces the function of ethical, transparent, and data-driven governance. As cities become increasingly complex in structure and in service delivery, predictive analytics offers the ability to create a framework to monitor more effectively, deploy resources more efficiently, and to develop community-based safety initiatives.

In summary, this dashboard provides more than just a technical answer for forecasting and ranking urban safety; it provides a framework for a strategic vision for preventing crime and building urban resilience - demonstrating how technology and data science can be leveraged to further public good.

References

Dataset Collection – via Kaggle

1. Crime in India State-wise data from 2001 is classified according to 40+factors. (75+ csv files)
2. Indian Crimes Dataset